

**SECOND PUBLIC EXAMINATION**

**Honour School of Physics Part B: 3 and 4 Year Courses**

**Honour School of Physics and Philosophy Part B**

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**B1: FLOWS, FLUCTUATIONS AND COMPLEXITY**

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**TRINITY TERM 2019**

**Friday, 14 June, 2.30 pm – 4.30 pm**

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

*A list of physical constants and conversion factors accompanies this paper.*

*The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.*

**Do NOT turn over until told that you may do so.**

**Page 2 contains physics formulae and data for this paper.**

**The questions start on page 3.**

The Navier–Stokes equation for viscous, incompressible, fluid flow under gravity is

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \frac{1}{\rho} \nabla p + g \mathbf{k} = \nu \nabla^2 \mathbf{u} ,$$

where  $\mathbf{u}$  is the fluid velocity,  $\rho$  the density,  $p$  the pressure,  $g$  the acceleration due to gravity,  $\mathbf{k}$  the vertical unit vector and  $\nu$  the kinematic viscosity.

The *Jacobian*  $\mathbf{J}$  of a dynamical system  $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$  is defined as  $J_{ij} = \frac{\partial f_i}{\partial x_j}$ , where  $f_i$  and  $x_j$  are the  $i^{\text{th}}$  and  $j^{\text{th}}$  components of  $\mathbf{f}(\mathbf{x})$  and  $\mathbf{x}$  respectively. The trace  $\tau$  of the Jacobian is equal to the sum of its eigenvalues, while the determinant  $\Delta$  is equal to their product.

For problems in atmospheric physics, unless otherwise stated,  $x$  and  $y$  refer to the horizontal eastward and northward directions and  $z$  the upward direction with corresponding velocities  $u$ ,  $v$  and  $w$ .

$$1000 \text{ mb} = 1000 \text{ hPa} = 10^5 \text{ Pa}.$$


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1. (a) Write down an expression for the vorticity  $\omega$  and give a physical interpretation. Write down expressions for the circulation as they relate to (i) the velocity field, and (ii) the vorticity. Explain the meaning of these expressions. [3]

(b) (i) Consider a fluid undergoing a solid body rotation with angular velocity  $\Omega$  in which the vorticity of the fluid  $\omega$  can be assumed uniform. Show that  $\omega$  is twice the rotational rate, using the expression for the circulation from (a). (ii) Consider a vortex line with a velocity field given by  $U_\theta(r) = A/r$ . Calculate the vorticity and the circulation. Comment on your results. [6]

Consider an incompressible, viscous and steady flow. The flow is sheared between two infinite co-axial and rotating cylinders, and therefore the flow is axisymmetric. The inner cylinder at  $r = r_1$  and the outer cylinder at  $r = r_2$  are rotating at angular rotational speeds of  $\Omega_1$  and  $\Omega_2$ , respectively. In cylindrical coordinates

$$\nabla^2 \mathbf{U} = \left( \nabla^2 U_r - \frac{U_r}{r^2} - \frac{2}{r^2} \frac{\partial U_\theta}{\partial \theta}, \nabla^2 U_\theta - \frac{U_\theta}{r^2} + \frac{2}{r^2} \frac{\partial U_r}{\partial \theta}, \nabla^2 U_z \right).$$

(c) Write down the continuity equation, and the radial and tangential components of the Navier–Stokes equations for the flow. You can neglect gravity. Show that the tangential velocity given by

$$U_\theta(r) = C_1 r + \frac{C_2}{r}$$

is a solution for the flow. What are the boundary conditions at  $r = r_1$  and  $r = r_2$ ? Show that  $C_1$  is given by

$$C_1 = \frac{\Omega_2 r_2^2 - \Omega_1 r_1^2}{r_2^2 - r_1^2},$$

and find  $C_2$ . Find the vorticity of the flow. Give a physical interpretation of  $C_1$ . Find the pressure as a function of  $r$ . [11]

(d) The approximations above are valid if the non-dimensional number

$$Re = \frac{\Omega_1 r_1 (r_2 - r_1)}{\nu},$$

where  $\nu$  is the kinematic viscosity, is less than 50. Explain the meaning of  $Re$ , and describe the expected behaviour of the flow as the parameters  $\Omega_1$ ,  $r_2 - r_1$ , and  $\nu$  are varied.

The inner cylinder with a radius of 0.25 m is rotating at 20 revolutions per minute, and the gap between the inner and outer cylinder is 5 mm. Assuming that fluid between the cylinders is filled with oil, with dynamic viscosity 0.02 Pa s, and density 900 kg m<sup>-3</sup>, and that  $C_2 = 0$ , calculate  $Re$ , find the pressure difference between the cylinders, and interpret your results. [5]

2. (a) Starting from the Navier–Stokes equation and neglecting gravity, show that

$$\frac{\partial \mathbf{u}}{\partial t} = -\nabla \left( \frac{p}{\rho} + \frac{1}{2} \mathbf{u}^2 \right) + \mathbf{u} \times \boldsymbol{\omega} + \nu \nabla^2 \mathbf{u},$$

where  $\mathbf{u}$  is the fluid velocity,  $\rho$  the density,  $p$  the pressure,  $\nu$  the kinematic viscosity, and  $\boldsymbol{\omega}$  the vorticity. Which additional assumption was necessary? [2]

[Hint: use the identity  $(\mathbf{u} \cdot \nabla) \mathbf{u} = \nabla (\frac{1}{2} \mathbf{u}^2) + \boldsymbol{\omega} \times \mathbf{u}$ .]

- (b) Assuming that the flow is irrotational and inviscid, show that

$$\frac{\partial \phi}{\partial t} + \frac{p}{\rho} + \frac{1}{2} \mathbf{u}^2 = C(t),$$

where  $\phi$  is the velocity potential, and  $C(t)$  is an arbitrary function of time. [3]

At  $t = 0$ , consider a spherical bubble with radius  $R_0$  at rest in an incompressible infinite fluid. At all times, we neglect gravity and the pressure inside the bubble ( $p_0 = 0$ ), and assume that the pressure at infinity,  $p_\infty$ , is constant and positive. The difference in pressure causes the bubble to collapse at  $t > 0$  such that  $R(t) < R_0$ .

- (c) Using continuity and the appropriate boundary condition, find the velocity of the flow, just outside the bubble, and show that the velocity potential is given by

$$\phi(r, t) = -\frac{R^2}{r} \frac{dR}{dt}.$$

[6]

- (d) (i) Determine the second order ordinary differential equation for the time evolution of the bubble's radius  $R(t)$ . (ii) Subsequently, show that the rate of change of  $R$  is given by

$$\frac{dR}{dt} = B \left( \frac{R_0^3}{R^3} - 1 \right)^{1/2}.$$

Find the expression for  $B$ .

[Hint for (ii): multiply the ODE found in (i) by  $2R^2 \frac{dR}{dt}$  and consider  $\frac{d}{dt} \left( \left( \frac{dR}{dt} \right)^2 R^3 \right)$ .] [11]

- (e) Deduce that the bubble collapses in the finite time

$$\Delta t = I \sqrt{\frac{3\rho}{2p_\infty}} R_0,$$

where

$$I = \int_0^1 \frac{dx}{\sqrt{x^{-3} - 1}} \approx 0.75.$$

Estimate  $\Delta t$  for a bubble in water of initial radius 1 mm and pressure  $10^5 \text{ N m}^{-2}$ . [3]

**3.** Consider a system with a mass attached to a spring. The mass  $m$  rests at the bottom of a tank and the spring is attached to one of the side walls. The force associated with a nonlinear spring is  $F_1 = -kx - \alpha x^3$ , where  $x$  is the displacement,  $k$  and  $\alpha$  are properties of the spring. The tank is filled with a fluid which exerts a drag force proportional to the velocity given by  $F_2 = -\beta v$ .

(a) Show that the system can be described by unforced Duffing equations:

$$\begin{aligned}\dot{x} &= y, \\ \dot{y} &= -ay - bx - cx^3.\end{aligned}$$

Give expressions for  $a$ ,  $b$  and  $c$ . [4]

(b) Assuming  $a = 0$  (no damping), and  $b = -c$ , calculate the fixed points, determine their stability, and classify them. Sketch a full phase portrait using  $b = -1$ . [6]

(c) Assuming  $a = 0$  (no damping), and considering all values of  $b$  and  $c$ , calculate all fixed points,  $x^*$ , determine their stability and classify them. Draw a bifurcation diagram for  $x^*$  as function of  $bc$ . Sketch a full phase portrait for  $b > 0$ . [11]

(d) Consider the following assumptions: weak damping with  $a^2 < 4|b|$ , the parameter  $b$  is negative  $b < 0$  and the coordinates of the fixed points are as in part (c). By using your findings from parts (a)–(c) and the assumptions stated here, describe and sketch the modifications to the phase portrait from part (b). Give a physical interpretation. [4]

4. Consider a polymer modeled as a three-dimensional freely jointed chain (FJC) of  $N$  rigid segments of length  $b$  and end-to-end vector  $\mathbf{r}$ , for which  $\langle \mathbf{r}^2 \rangle = Nb^2$ .

(a) If the probability for the projection of  $\mathbf{r}$  along  $x$  is described by

$$P_x(x, N) = \left( \frac{3}{2\pi Nb^2} \right)^{\frac{1}{2}} \exp \left( \frac{-3x^2}{2Nb^2} \right)$$

and the projections along the  $y$  and  $z$  dimensions follow the same functional form, calculate  $P(\mathbf{r}, N)$  for the chain. Plot  $P(x, N)$ , discuss how it changes as a function of  $N$ , and identify similarities between FJC conformations and Brownian motion. [6]

(b) Show that the entropic energy for the extension of a FJC to length  $r$  is given by

$$F(\mathbf{r}) = d + c\mathbf{r}^2$$

where  $d$  and  $c$  are constants. Provide an expression for  $c$ , and explain why a FJC functions as an entropic spring. How will the entropic spring stiffness change with increased temperature? [4]

(c) Assume that force  $f$  is applied to the ends to the FJC polymer to extend its conformation. Derive an expression for the partition function of the FJC under force, and show that the mean end-to-end distance  $z$  is described by

$$z = Nb \left( \coth(a) - \frac{1}{a} \right)$$

where  $a = fb/(k_B T)$ . Derive what the force-extension curve would look like in the low-force limit for the FJC, where you can assume that  $\coth(a) \approx a/3 + 1/a$ . [6]

(d) Consider a linear double-stranded DNA with basepair (bp) steps of 0.34 nm, and a persistence length  $P = 150$  bp. If the DNA is treated as a FJC, what is the probability  $p(\lambda)$  of the DNA ends coming in contact? What is the dependence of  $p(\lambda)$  on  $N$ ? Show that the entropic penalty for conformations with DNA ends in contact is given by  $\Delta S = k_B(C - \frac{3}{2} \log N)$ , where  $C$  is a constant. [5]

(e) Assume that the DNA ends come in contact by forming DNA loops, wherein the elastic energy for bending the DNA is  $\Delta G_{\text{bend}} = (\pi P/R)k_B T$ , and  $R$  is the radius of curvature of the loop. Assuming that the elastic and entropic terms are additive, write an expression for the looping free energy  $\Delta G_{\text{loop}}$ , plot and discuss the dependence of  $\Delta G_{\text{loop}}$  on DNA length, and calculate the DNA length where  $\Delta G_{\text{loop}}$  reaches a minimum. [4]